

Due before our final exam. You must show work to receive credit. Each problem is worth 4 out of 100 points.

Decide whether each of the following is statement. If it is a statement, decide whether or not it is compound.

1. JCSU is a university, and I am a student.

2. Is Dr Ashe an elephant?

Write a negation of each statement.

3. $y \leq 5$

4. Jane lives in the dorm.

Fill in the following truth tables.

5.

p	q	$\sim (p \vee q)$
T	T	
T	F	
F	T	
F	F	

Let p represent a false statement and q represent a true statement. Find the truth value of each compound statement.

6. $p \vee \sim q$

7. $\sim (p \vee \sim q)$

Let p represent a true statement, and q and r represent false statements. Find the truth value of each compound statement.

8. $q \vee \sim r$

9. $(\sim p \vee q) \wedge \sim r$

Let p be the statement " $2 > y$ ", q be statement " $8 \leq 6$," and r be the statement " $19 \leq 19$." Find the truth value of each compound statement.

10. $\sim p \vee \sim r$

11. $(\sim r \vee q) \vee \sim p$

Let $A = \{2, 4, 6, 10\}$, $B = \{2, 10\}$, and $C = \{2, 4, 6, 10\}$. Decide whether the following are true or false.

12. $B \subset A$

13. $A \subseteq B$

Let $E = \{a, b, c\}$.

14. How many subsets does E have? List them.

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$, $B = \{2, 3, 4, 5, 6\}$, and $C = \{1, 2, 3, 8, 9\}$. List the members of each set, using set braces.

15. $A \cup B$

16. $A \cup (B \cap C)$

17. $A' \cap B$

Use a Venn diagram to answer the following question.

18. Of the students in your class, 10 have sisters, 15 have brothers, 5 have a brother and a sister, and 7 have neither a brother or a sister.

a. How many students have a brother or a sister?

b. How many students are in the class?

A fair six sided dice is thrown. Find the probability of each event.

19. Getting a 3.

20. Getting a 3 or 6.

21. Not getting a 3.

22. Not getting a 3 or 6.

A fair coin is tossed three times in a row. Find the probability of each event.

23. Getting 3 heads in a row.

24. Getting at least two tails in a row (equivalently getting TTH or TTT).